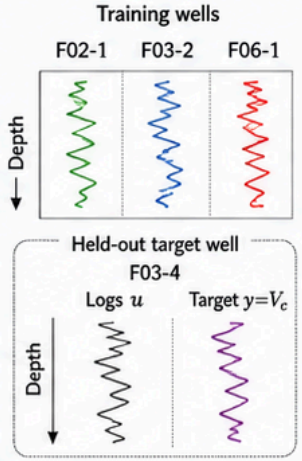
Experimental workflow for CLP-CSGM evaluation

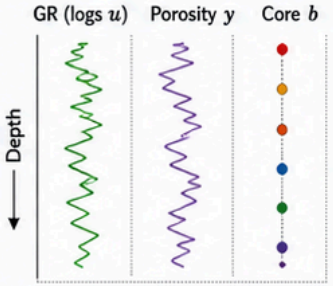
1. Data and experimental regimes

A Cross-well low-data benchmark



- Training wells: F02-1, F03-2, F06-1
- Held-out target well: F03-4
- Dense logs u
- Target property $y = V_c$
- Windowing + normalization
- Low-data settings (step / reduced training density)
- Sparse measurements generated by subsampling the target profile

B Real-well F03 benchmark



- Well: F03-4
- Dense logs u (GR-only stress test)
- Target property: porosity
- Sparse core measurements b
- Measurement ratio ρ varied

2. Model construction and inference

Train autoencoder (E, G) on training target windows y

Frozen decoder G

Encoded latents $z = E(y)$

Fit conditional latent prior $h : u \to z_0$ (Ridge)

Target-well inputs: dense logs u^* + sparse measurements b^*

Latent anchor $z_0 = h(u^*)$

CLP-CSGM Ridge

$$\hat{z} = \arg\min_z \| MG(z) - b^* \|_2^2 + \lambda \| z - z_0 \|_2^2$$

Reconstruction: $\hat{y} = G(\hat{z})$

Direct baselines

- ML only
- MLP $[u, b]$
- PCA $[u, b]$
- AE $[u, b]$

Ablations

- Prior-only: $G(h(u))$
- Measurement-only CSGM
- Prior class sensitivity: Ridge vs MLP

3. Experimental sweep and evaluation

Sweep measurement ratio ρ in $\{0.20, 0.30, 0.40, 0.50, 0.60\}$

Run multiple seeds / settings

Metrics: RMSE, MAE, relative L_2

Diagnostics: parity plots, profile overlays, runtime vs error

Aggregate tables and figures

4. Main empirical findings

Cross-well low-data:
CLP-CSGM Ridge outperforms AE $[u, b]$ across the evaluated grid.

Real-well F03 GR-only:
strongest in sparse regimes (low ρ); AE $[u, b]$ may catch up at higher ρ.

Key interpretation:
sparse target measurements are most effective as test-time measurement constraints over a conditional generative latent prior.